

## **Project Report**

### **Amazon Delivery Time Prediction**

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#### **Introduction**

Efficient and accurate delivery time prediction is crucial in modern e-commerce logistics. With increasing customer expectations and dynamic delivery conditions such as traffic, weather, and geographic distance, businesses must rely on data-driven approaches to estimate delivery times. This project focuses on predicting delivery times using machine learning models trained on real-world e-commerce delivery data.

#### **Problem Statement**

The objective of this project is to develop a predictive system that estimates delivery times based on various factors such as product category, agent characteristics, weather, traffic, and geographic distance.

#### **Business Use Cases**

- Enhanced Delivery Logistics
- Dynamic Traffic & Weather Adjustments
- Agent Performance Evaluation
- Operational Optimization

#### **Dataset Overview**

The dataset `amazon_delivery.csv` contains information about orders, agents, delivery conditions, and delivery durations.

#### **Methodology**

- i. Data Preparation
- ii. Data Cleaning
- iii. Exploratory Data Analysis
- iv. Feature Engineering
- v. Model Development

#### **Exploratory Data Analysis**

Key insights include correlation between distance, weather, traffic, and delivery time.

## **Feature Engineering**

Geospatial distance and time-based variables (hour, day, etc.) were extracted.

## **Model Development**

Models trained:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Gradient Boosting performed best.

## **MLflow Tracking**

All model parameters, metrics, and versions were logged for comparison.

## **Streamlit Application**

A user interface allows users to input conditions and get predicted delivery times.

## **Deployment**

Model deployed using Streamlit frontend.

## **Results & Insights**

Traffic, weather, and distance were influential predictors.

## **Conclusion**

The project demonstrates a full ML pipeline from preprocessing to deployment.

## **Future Work**

Integration of real-time traffic APIs, deep learning models, and containerized deployment.

## **Technologies Used**

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, MLflow, Streamlit.